

Adwait B. Naravane

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WORK EXPERIENCE

Doctoral Researcher, UGent

October 2024 - Present

FWO Doctoral Fellow

I am currently working on the development and the implementation of Tensor Network Algorithms to study Lattice Gauge theories, Statistical Mechanics and Quantum Many Body systems. I am working on the development on TNRKit.jl. an open-source julia library based off TensorKit.jl. The software is designed to perform Tensor network renormalisation simulations for two and three dimensions and has applications in the aforementioned fields. I am also working on the development of PEPS algorithms for two dimensional quantum systems through PEPSKit.jl. I am also a Master's thesis guide and a Graduate teaching assistant for advanced Physics courses.

Supervisor: Prof. Jutho Haegeman, UGent

Master's thesis student, TIFR

September 2023 - May 2024

Worked on implementing Tensor network algorithms to study Quantum many body systems on two-dimensional lattices. The algorithms include Density Matrix Renormalisation Group (DMRG) and Projected entangled pair states (PEPS) which were implemented on the TIFR cluster using the Julia Language.

Supervisor: Prof. Nilmani Mathur, Prof. Kedar Damle, TIFR

Undergraduate Research Intern, RIKEN CCS

June 2023 - August 2023

I joined the Computational material sciences team at the RIKEN Center for Computational Sciences at Kobe, Japan. I worked on implementing Tensor network algorithms associated with Matrix product states (MPS) to study two dimensional Fermionic systems. The algorithms were implemented using Python, Julia and MPI parallel computing on the Fugaku supercomputer.

Supervisor: Dr. Seiji Yunoki, RIKEN

Undergraduate Teaching assistant

January 2023 - May 2023

Teaching assistant for the Computational Physics Lab. Held tutorials on numerical methods and helped students with learning basic numerical methods used to study Quantum Physics.

Supervisor: Prof. Rangeet Bhattacharya, IISER-K

Undergraduate Researcher, TIFR

May 2022 - December 2022

Studied various lattice field theories such as Ising, XY models, and Scalar Field theory under Prof. Nilmani Mathur at TIFR. I worked on implementing a program that generates uncorrelated spin configurations for these models in Python and Julia using Quantum Monte Carlo methods. I worked on building neural networks like Conditional Variational Autoencoders using Tensorflow to detect & study phase transitions in these models. The results of this research have led to a paper mentioned below.

Supervisor: Prof. Nilmani Mathur, TIFR

EDUCATION

October 2024 - present PhD student at **Ghent University**

Aug 2019 - June 2024 BS-MS at **IISER-Kolkata** (GPA: 8.76/10)

June 2017 - May 2019 Class 12 Board at **D.G.Ruparel College, Mumbai** (GPA: 9.2/10)

ADDITIONAL INFORMATION

Languages Python, Julia

Tools & other Software Message Passing Interface (MPI), QuSpin, Itensor, TensorKit, Quantum Espresso, Tensorflow, Numpy, Pandas

PUBLICATIONS

- Semi-supervised learning of order parameter in 2D Ising and XY models using Conditional Variational Autoencoders
Adwait Naravane, Nilmani Mathur, e-Print: 2306.16822 [cond-mat.stat-mech]
- Global Tensor Network Renormalization for 2D Quantum systems: A new window to probe universal data from thermal transitions
A Ueda, S De Meyer, A Naravane, V Vanthilt, F Verstraete, e-Print: 2508.05406 [cond-mat.str-el]
- Deconfinement from Thermal Tensor Networks: Universal CFT signature in (2+1)-dimensional Z_N lattice gauge theory
Adwait Naravane, Yuto Sugimoto, Shinichiro Akiyama, Jutho Haegeman, Atsushi Ueda, e-Print: 2602.13124 [hep-th]